EXPERIMENT

PROOFING TIME AND TEMPERATURE RECOMMENDATIONS

Our master recipes for each kind of pizza dough include times and temperatures for proofing. These conditions will bring out the best in the dough and will consistently deliver a dough with the specific key characteristics for each pizza (see page 3:3). Be aware that these times are guidelines, however, and keep your eye on your dough as it proofs.

In some pizzerias or homes, however, it may be difficult or impossible to create the recommended conditions. For example, air conditioning or ample refrigerator storage space may not be available. You might have tight time constraints or, on the flip side, there are cases when slowing down dough development may be more practical. In the latter scenario, cold-proofed doughs that are dormant can be held relatively stable until the pizza maker is ready to use them. Note that weaker yeast activity and slower dough development will continue even when refrigerated.

We experimented to find out whether each of our master doughs can proof at different temperatures and, if so, how long that takes and whether they need to be handled differently. Using the following table, you'll be able to delay, hold, or speed up the dough development process depending on the conditions.

We cold-proofed New York and artisan pizza doughs for 7 days at 4° C / 39° F. These doughs turned out to be relatively stable and delivered a positive result after the first day and up to 4 days. After 5 days, the dough declined in yeast activity, and the pizza began to lose flavor and structure.

We tried proofing our master thin-crust pizza dough at typical room-temperature proofing conditions and then baked pizzas after 1, 2, 3, 4, 6, and 24 hours. The 2- and 3-hour proofing times produced pizzas with the best attributes, which is what we have included in our master recipe on page 110 (you can opt to cold-proof the dough after this point if you'd like; it makes the dough easier to handle). At 6 hours the dough at room temperature was overproofed, and the pizza quality started to decline.

Because many types of bread are warm proofed, we tried this with pizza. When warm-proofing, the majority of the bread-like pizzas are typically proofed for 2 hours at 27° C / 80° F. Our focaccia, New York square, and high-hydration al taglio doughs were also tested at 1-, 3-, and 4-hour proof times to see how they compared. For focaccia, the 1-hour proofed dough gave a tighter crumb with a volume that was lower than our master focaccia by a statistically significant margin. With an increase in proofing time, the focaccia dough became slacker and gassier, with gas bubbles becoming more noticeable on top of the baked pizza. The flavor of the 4-hour focaccia was very sour and the texture became softer.

For our New York square dough, the 3-hour proofed pizza resulted in a small and uneven crumb, while a drastic reduction in volume was observed after 4 hours. The high-hydration al taglio dough was malleable and supple at all of the times we tested. The volume of the 1-hour proofed pizza was inferior, however, and fell out of the statistical range of volumes (see page 396), confirming that this proof time is insufficient. The 2-, 3-, and 4-hour pizzas had an even golden color, a good flavor, an expected chew, and a crispy crumb. We concluded that these doughs can tolerate a range of 2–4 hours at warm-proofing temperatures; in our master recipes, we recommend proofing them for 3 hours.

We should caution, however, that alternative proof temperatures listed below could also lead to different results. For our cold-proofed doughs, for example, refrigeration does more than just slow down the yeast activity. During cold-proofing, the dough itself continues to develop flavor attributes that are critical to the outcome of the pizza. Because of this, we strongly recommend cold-proofing for New York and artisan pizza doughs. Using room-temperature or warm-proofing in these doughs leads to underdeveloped flavors and a lack of crispiness. Although some proofing temperatures can lead to acceptable results without major alterations to the final products, pizza makers should use the recipe's recommended proof temperatures whenever possible.

OUR PROOFING TIME AND TEMPERATURE RECOMMENDATIONS

We proofed each of our master doughs in the three conditions listed below to see how each performed. Cold- and room-temperature proofing made thin-crust and medium-crust pizzas that were crisp and flavorful. Room-temperature and warm-proofing provided the best results for pan-baked pizzas, which were crisp and light with a good chew. For thin-crust pizzas, you can roll out cold-proofed dough without tempering. For medium-crust pizzas, cold-proofed dough needs to temper for 2 hours at room temperature before using.

Cold-proofing (4° C / 39° F)	Room temperature (21° C / 70° F)	Warm-proofing (27° C / 80° F)
Thin-crust pizza dough (24 h)	Thin-crust pizza dough (2 h)	Focaccia dough (2 h)
Brazilian thin-crust pizza dough (24 h)	Brazilian thin-crust pizza dough (1½ h)	New York square pizza dough (2 h)
Deep-dish pizza dough (24 h)	Deep-dish pizza dough (2–4 h)	High-hydration al taglio pizza dough (2 h)
Neapolitan pizza dough with poolish (48 h)	Neapolitan pizza dough (20–24 h)	Detroit-style pizza dough (2½ h)
New York pizza dough (48 h)	Focaccia dough (3 h)	
Artisan pizza dough (48 h)	New York square pizza dough (3 h)	
	High-hydration al taglio pizza dough (3 h)	
	Detroit-style pizza dough (3 h)	

We tried proofing our focaccia and high-hydration al taglio pizza doughs, with yeast or levain or both, in the wine refrigerator at 13° C / 55° F because that was one of the tricks we learned in *Modernist Bread* to improve the results of doughs made with levain. Unfortunately, the results were less than stellar and we don't recommend it for these pizzas.

We recommend tempering cold-proofed dough for 2 h (or for however long it takes for the dough to come up to 13° C / 55° F) before baking it.

Check the dough to see if it is fully proofed and ready to bake. Use your finger to gently press the exposed surface of the dough for 2 seconds. The pressure should leave a small dent in the dough. It will slowly spring back, but the indentation will remain clearly visible.

Proper proof

Overproofed

Dough CPR performed

CALLING PROOF

Most people can't call proof accurately because their intuition is to look for a doubling in the height or diameter of the piece of dough. Even if you can judge volume accurately (we use a laser scanner to do so), doubling doesn't reflect fermentation time accurately. Nor does it give you the information you need before baking. Are the bubbles in the dough fully inflated but not ruptured? Has fermentation progressed enough to fully develop the flavor? Does the dough retain enough elasticity to accommodate oven spring without risking damage to the texture of the crumb?

Calling proof is hard to teach. Even experienced pizzaioli find it can be one of their most difficult tasks. Success comes from a hands-on familiarity with the feel of a properly proofed dough and an understanding of its fermentation process. It's important to think about the yeast and what it's doing long before the shaping and final proofing stages. Fermentation starts practically at the instant the flour and yeast become hydrated. If the dough was bulk-fermented for too long or sat on the worktable for an extended period before dividing or shaping, it will either proof very quickly once it's shaped or may even overproof before that. You can usually extend the final proof of the dough by refrigerating it once it's been shaped. The type of dough will dictate its adaptability to such factors.

The right moment for calling proof is when there's an optimal relationship between the amount of gas in the dough and the ideal gas-retention properties of the gluten network. Keep in mind that more gas will be produced during oven spring, so the gluten should still be strong enough to retain the gas and ensure proper volume and crumb structure. With experience, you'll be able to simply use touch to determine whether the dough is ready to bake, but few can call proof accurately on their first few doughs.

To further cloud the issue, some bakers slightly overproof their doughs intentionally. That's because it produces a much more open crumb structure and better flavor than just barely proofing a dough. An underproofed dough will lack flavor development and will likely have a tighter crumb because of insufficient fermentation. Note that if you err, it's safer to do so on the side of overproofing. If you've cold-proofed your dough for a long period—say, 24–48 hours—it isn't exactly necessary to call proof. By that point, it's likely ready to be pulled out, tempered for 2 hours, shaped, and then baked.

No kitchen gadget can simplify calling proof. But if you're adventurous and willing to practice on a few balls of dough, you can develop the skill through trial and error. Try the following simple experiments to get a better idea of how to call proof. Once the dough has been left to proof for the time we recommend, take a close look at it. Has its volume increased since it was shaped? Does it look bubbly and gassy? Does it jiggle when you poke it or move the tub? (It should.) Perform the fingertip test (see the next page) and bake the dough when it seems properly proofed. If the pizza turns out successfully, use that fingertip test as a guide for future balls of dough. Remember what that dough felt and looked like. Don't underestimate the power of muscle memory.

If the dough has overproofed, you'll see telltale signs. It will look like a balloon in the throes of deflating. You'll see wrinkles where there were bubbles of carbon dioxide. It will smell more like alcohol than yeast. When baked, the pizza may have a pale crust and a tight crumb, and you're likely to taste alcohol.

At the other end, dough that's underproofed will spring back quickly if you perform the fingertip test. When baked, the dough could rip and tear on the surface, and it's likely to become misshapen during baking. You'll see how flat the dough stays. This may not be a big deal for thin-crust, Brazilian thin-crust, and deep-dish pizzas, but for other styles, a super flat pizza is a telltale sign of underproofing. It's an instructive reminder of why dough needs time not only to proof but also to relax the gluten and tension created by shaping.

The Fingertip Test

In principle, there ought to be some scientific way to determine when proofing is complete. You could track the gradual decline in pH as fermentation progresses, for example, or measure the increase in the dough's volume, or quantify changes to its firmness and elasticity. We have tried all these in our lab. Our conclusion: no. Because so many things vary from one dough recipe to the next, no simple technical method can be used to accurately call proof for all kinds of dough. With enough trial and error, it's probably feasible to find values for a volume increase, drop in pH, or some other precise measure that signals the end of proofing for a specific recipe. But what most pizza makers need is an easy test that works for most recipes.

The fingertip test fits that bill. It's not a scientific method, and you'll have to practice to become proficient. But the good news is, you'll need only

one tool—an index finger. And although we don't all feel proofed dough the same way, the test is good enough for making an educated assessment of whether the dough is ready to move on to baking. To perform the fingertip test, use your finger to gently press the exposed surface of the dough for 2 seconds. The pressure should leave a small dent in the dough. It will slowly spring back, but the indentation will remain clearly visible. The steps, illustrated at right, are fast and straightforward and can be done repeatedly without harming the dough. The fingertip test works reasonably well because it integrates several different factors that change during proofing, including the water content of the dough, its gluten development, bubble integrity, and the amount of captured gas.

Dough CPR

It happens to the best of us. You wait for your dough to proof so you can bake it and then, somehow, you lose track of time and it overproofs. You may even bake the overproofed dough, hoping it will magically return to life; instead, you end up with a pale flat pizza that smells like an old beer.

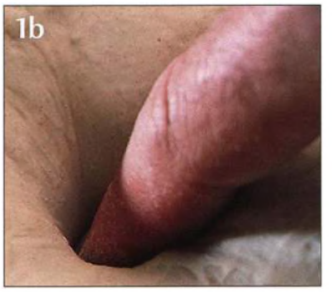
Conventional wisdom holds that overproofed doughs are irretrievably damaged and should be thrown away. Our experiments found just the opposite. In fact, we were able to resuscitate the same batch of dough up to 10 times before it suffered any serious loss in quality. The method is so simple that you'll want to cry for every batch of overproofed dough you've thrown out.

The first step is to confirm that your dough really is overproofed. It may just have excess gas rather than something more dire. During proofing, the size of the bubbles in the dough increase; as a result, the cell walls between the bubbles become thinner. Those walls undergo a process called shear hardening that makes them more elastic. At a point far beyond that, the cell walls rupture, and the bubbles coalesce. The gas stays in the dough, but there are fewer (though larger) cells. Overproofing involves an irreversible thinning and possible tearing of the cell walls during baking. It usually takes a while for this to actually occur, though.

Perform the fingertip test to determine the dough's condition: if the dent you've created with your finger doesn't spring back at all, the dough is probably overproofed. The bubble walls in overproofed doughs are stretched too thin, so they are mechanically unable to spring back. The smell of alcohol is also a sign. This telltale scent was key as we developed our method for determining and rescuing overproofed dough.

Because overproofing involves the thinning cell walls, along with excess carbon dioxide and alcohol, the trick to reviving your dough hinges on releasing the gas and alcohol and reinforcing the cell walls. Just degas the dough and reshape it. That's it.

This method works especially well for Neapolitan dough; it also works well for thin-crust, Brazilian thin-crust, New York, artisan, deep-dish, and Detroit-style pizza doughs. The key is to let the dough relax after reshaping it, giving it some time before shaping and baking. Reshaping produces a lot of tension in the dough and can make it very hard to stretch open without ripping. For our pizzas mentioned above, we wait 3–4 hours before shaping (see pages 3:12 and 3:66 for the shaping methods for these pizzas).



HOW TO Perform Dough CPR

1 Perform the fingertip test to make sure your dough is overproofed; the dent you make by gently poking the dough should be permanent (see the images in the margin above).

2 Remove the dough from the proofing vessel.

3 Degas the dough by pressing down firmly on it. Use the same pressure as you would for shaping.

4 Shape the dough into a ball, and return it to its proofing container. Cover well and let the dough relax for 3–4 h at room temperature.